Social learning by type-token frequency - analysis

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Abstract

Abstract (to be written). Argument structure - See below. Analysis structure - 1. Descriptives of difference scores and tests against zero. 2. GLMMs, with reference to ANOVA. Discuss quorum sensing [Sumpter and Pratt, 2009] Discuss moral repetition effect, and tolerance principle and submit to psych sci. In intro. Different formal perspectives suggest that mere frequency should not be relevant, From a learning perspective, [Explanation of tolerance principle; argue later that threshold is 8/ln(8) = 3.8, but that we did not vary this experimentally as it's too long; general explanation of tolerance principle-When having a productive rule is faster than listing all items as exception]. From a representational perspective, dispositions. intrinsic properties of banana; bayesian argument. From a social perspective, majority. Whatever the moral frequency guy argues.

We excluded 0 participants because they failed the attention test.

We next analyze the SD and combine the results with the data frame for the main experiment.

Table 1: Number of participants in each group of the SD3 scales. Note that this applies to city students only

	N			**Percentage**		
scale	high	low	NA_	high	low	NA_
Machiavellianism	23	24	5	0.442	0.462	0.096
Narcissism	26	26	0	0.500	0.500	0.000
Psychopathy	25	24	3	0.481	0.462	0.058

1 Summary

1.1 References for the moral/conventional distinction.

[Machery and Stich, 2022, Hauser et al., 2010]

1.2 Ratings

1.2.1 Prevalence

We observed a main effect of valency, suggesting that good actions were judged better than neutral actions, which were judged better than neutral actions. VERIFY THIS. GLMER: good > neutral = bad

We also observed a main effect of Action Status, suggesting that single occurence actions were judged as less prevalent than low frequency actions, which we judged as less prevalent than high type frequency actions.

The interaction was not significant.

1.2.2 Third party acceptability

We observed a main effect of valency, suggesting that good actions were judged better than neutral actions, which were judged better than neutral actions.

We also observed a main effect of Action Status, suggesting that single occurrence actions were judged as worse than low frequency actions, which we judged as worse than high type frequency actions.

These factors did not interact.

1.2.3 First party acceptability

We observed a main effect of valency, suggesting that good actions were judged better than neutral actions, which were judged better than neutral actions. VERIFY THIS. GLMER: good > neutral = bad

We also observed a main effect of Action Status, suggesting that single occurence actions were judged as worse than low frequency actions, which we judged as worse than high type frequency actions.

These factors interacted. The effect of Action Status was only significant for neutral and good action, where the effect size for neutral actions was twice as high as for good actions.

1.3 Difference scores

1.3.1 Prevalence

The difference score comparing the high/low type frequency condition against the single occurrence condition yielded a significant main effect of Action Status, suggesting that the difference score was greater for the high type frequency condition than for the low type frequency condition. While the main effect of valency did not reach significance, the the interaction reached significance. The effect size associated with Action Status was much larger for neutral scenarios.

The difference score comparing the high and the low typ frequency condition revealed a significant main effect of Valency, suggesting that that the difference score did not differ between bad and neutral actions, but between neutral and good actions.

1.3.2 Third party acceptability

The difference score comparing the high/low type frequency condition against the single occurrence condition yielded a significant main effect of Action Status, suggesting that the difference score was greater for the high type frequency condition than for the low type frequency condition. Neither the main effect of valency nor the interaction reached significance.

Following up this interaction, we found that the main effect of Valency was only significant for high type frequency actions, but not for low frequency actions. Further, the effect size association with Action Status was much larger for neutral scenarios.

The difference score comparing the high and the low typ frequency condition revealed a marginal main effect of Valency, suggesting that that the difference score did not differ between bad and neutral actions, but differed marginally between neutral and good actions.

1.3.3 First party acceptability

The difference score comparing the high/low type frequency condition against the single occurrence condition yielded a significant main effect of Action Status, suggesting that the difference score was greater for the high type frequency condition than for the low type frequency condition. The main effect of valency reached significance, as did the interaction.

Following up this interaction, we found that the main effect of Valency was only significant for high type frequency actions, but not for low frequency actions. Further, the effect size association with Action Status was much larger for neutral scenarios, and reached significance only for these scenarios.

The difference score comparing the high and the low typ frequency condition revealed a marginal main effect of Valency, suggesting that that the difference score differed marginally between bad and neutral actions, but differed significantly between neutral and good actions.

2 Raw ratings

2.1 Ratings

Table 2: Descriptives for the untransformed as well as the binary ratings. The p value reflects a Wilcoxon test against the scale midpoint of 2.5 or the chance level of .5.

		Raw ratings				**Binary ratings**			
	N	M	SE	Cohen's d	p	M	SE	Cohen's d	p
city - bad - prevalence									
highTypeFreq	17	3.53	0.104	8.45	0.000	0.936	0.025	9.359	0.000
lowTypeFreq	17	2.75	0.164	4.18	0.181	0.534	0.104	1.279	0.792
single	17	2.09	0.115	4.53	0.005	0.235	0.073	0.802	0.005
city - neutral - prevalence									
highTypeFreq	17	3.51	0.117	7.48	0.000	0.882	0.040	5.462	0.000
lowTypeFreq	17	2.50	0.091	6.88	0.875	0.431	0.063	1.723	0.307
single	17	1.96	0.114	4.29	0.001	0.206	0.045	1.147	0.001
city - good - prevalence									
highTypeFreq	18	3.75	0.071	12.91	0.000	0.977	0.012	20.404	0.000
lowTypeFreq	18	3.29	0.134	5.97	0.000	0.819	0.066	3.001	0.002
single	18	2.70	0.149	4.39	0.303	0.546	0.086	1.539	0.614
city - bad - third.party.acceptablity									
highTypeFreq	17	2.71	0.146	4.63	0.192	0.637	0.077	2.072	0.070
lowTypeFreq	17	1.89	0.121	3.88	0.001	0.211	0.061	0.867	0.003
single	17	1.66	0.115	3.59	0.000	0.103	0.039	0.654	0.000
city - neutral - third.party.acceptability									
highTypeFreq	17	3.46	0.095	9.14	0.000	0.882	0.037	5.983	0.000
lowTypeFreq	17	2.72	0.098	6.92	0.038	0.569	0.054	2.645	0.218
single	17	2.32	0.119	4.89	0.097	0.382	0.055	1.749	0.032
city - good - third.party.acceptability									
highTypeFreq	18	3.80	0.060	15.42	0.000	0.977	0.012	20.404	0.000
lowTypeFreq	18	3.45	0.117	7.16	0.000	0.894	0.041	5.224	0.000
single	18	3.00	0.155	4.70	0.007	0.676	0.078	2.099	0.043
city - bad - first.party.acceptablity									
highTypeFreq	17	1.61	0.094	4.30	0.000	0.123	0.030	1.036	0.000
lowTypeFreq	17	1.57	0.099	3.98	0.000	0.098	0.028	0.882	0.000
single	17	1.54	0.097	4.00	0.000	0.088	0.036	0.605	0.000
city - neutral - first.party.acceptablity									
highTypeFreq	17	2.81	0.106	6.64	0.004	0.603	0.050	3.033	0.056
lowTypeFreq	17	2.59	0.105	6.19	0.705	0.539	0.047	2.889	0.693
single	17	2.47	0.124	4.96	0.571	0.500	0.050	2.502	0.752
city - good - first.party.acceptablity									
highTypeFreq	18	3.69	0.070	12.86	0.000	0.940	0.022	10.500	0.000
lowTypeFreq	18	3.68	0.075	11.85	0.000	0.944	0.023	9.963	0.000
single	18	3.63	0.083	10.57	0.000	0.931	0.024	9.302	0.000
testable - bad - prevalence									
highTypeFreq	41	3.48	0.098	5.59	0.000	0.888	0.034	4.155	0.000
lowTypeFreq	41	2.96	0.118	3.98	0.001	0.650	0.054	1.901	0.007
single	41	2.27	0.070	5.10	0.005	0.325	0.052	0.993	0.002

Table 2: Descriptives for the untransformed as well as the binary ratings. The p value reflects a Wilcoxon test against the scale midpoint of 2.5 or the chance level of .5. *(continued)*

		Raw ratings				Binary ratings			
	N	M	SE	Cohen's d	p	M	SE	Cohen's d	p
testable - neutral - prevalence									
highTypeFreq	39	3.66	0.078	7.64	0.000	0.923	0.025	5.882	0.000
lowTypeFreq	39	2.90	0.106	4.46	0.001	0.628	0.059	1.739	0.026
single	39	2.33	0.095	3.97	0.063	0.368	0.058	1.032	0.009
testable - good - prevalence									
highTypeFreq	39	3.79	0.053	11.66	0.000	0.968	0.014	11.151	0.000
lowTypeFreq	39	3.43	0.073	7.58	0.000	0.923	0.025	5.971	0.000
single	39	2.96	0.099	4.86	0.000	0.682	0.053	2.083	0.001
testable - bad - third.party.acceptability									
highTypeFreq	41	2.95	0.129	3.63	0.002	0.720	0.052	2.197	0.000
lowTypeFreq	41	2.46	0.133	2.92	0.619	0.459	0.058	1.246	0.519
single	41	1.86	0.091	3.22	0.000	0.215	0.045	0.757	0.000
testable - neutral - third.party.acceptability									
highTypeFreq	39	3.58	0.077	7.53	0.000	0.895	0.028	5.149	0.000
lowTypeFreq	39	3.11	0.095	5.32	0.000	0.754	0.041	2.958	0.000
single	39	2.77	0.088	5.08	0.005	0.635	0.047	2.210	0.018
testable - good - third.party.acceptability									
highTypeFreq	39	3.84	0.050	12.52	0.000	0.979	0.014	11.528	0.000
lowTypeFreq	39	3.65	0.060	9.87	0.000	0.959	0.017	9.339	0.000
single	39	3.34	0.077	7.05	0.000	0.861	0.031	4.542	0.000
testable - bad - first.party.acceptablity									
highTypeFreq	41	1.35	0.054	3.98	0.000	0.061	0.016	0.607	0.000
lowTypeFreq	41	1.32	0.060	3.51	0.000	0.063	0.023	0.441	0.000
single	41	1.30	0.057	3.60	0.000	0.065	0.024	0.425	0.000
testable - neutral - first.party.acceptablity									
highTypeFreq	39	3.10	0.071	7.08	0.000	0.699	0.030	3.798	0.000
lowTypeFreq	39	2.97	0.077	6.24	0.000	0.669	0.032	3.390	0.000
single	39	2.90	0.082	5.70	0.000	0.671	0.035	3.103	0.000
testable - good - first.party.acceptablity									
highTypeFreq	39	3.79	0.043	14.47	0.000	0.966	0.010	16.206	0.000
lowTypeFreq	39	3.72	0.051	11.83	0.000	0.953	0.014	11.204	0.000
single	39	3.68	0.053	11.33	0.000	0.953	0.013	11.498	0.000

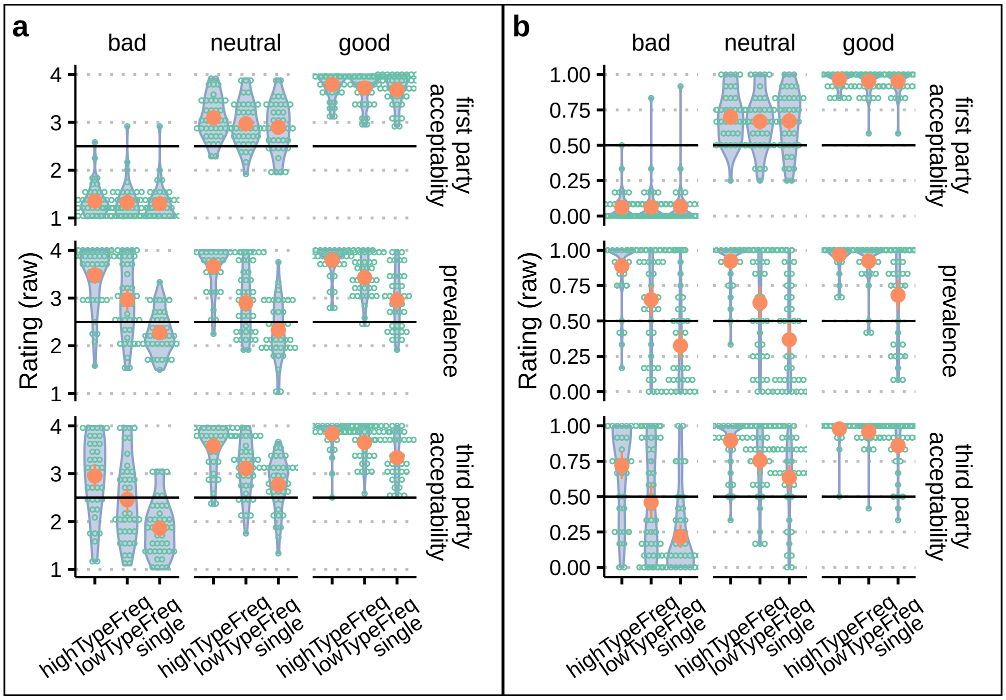

Figure 1: Ratings for testable participants. NOTE THAT WE HAVE SPECIFIC PLOTS FOR BINARY RATINGS BELOW. (a) Raw ratings. (b) Binary ratings

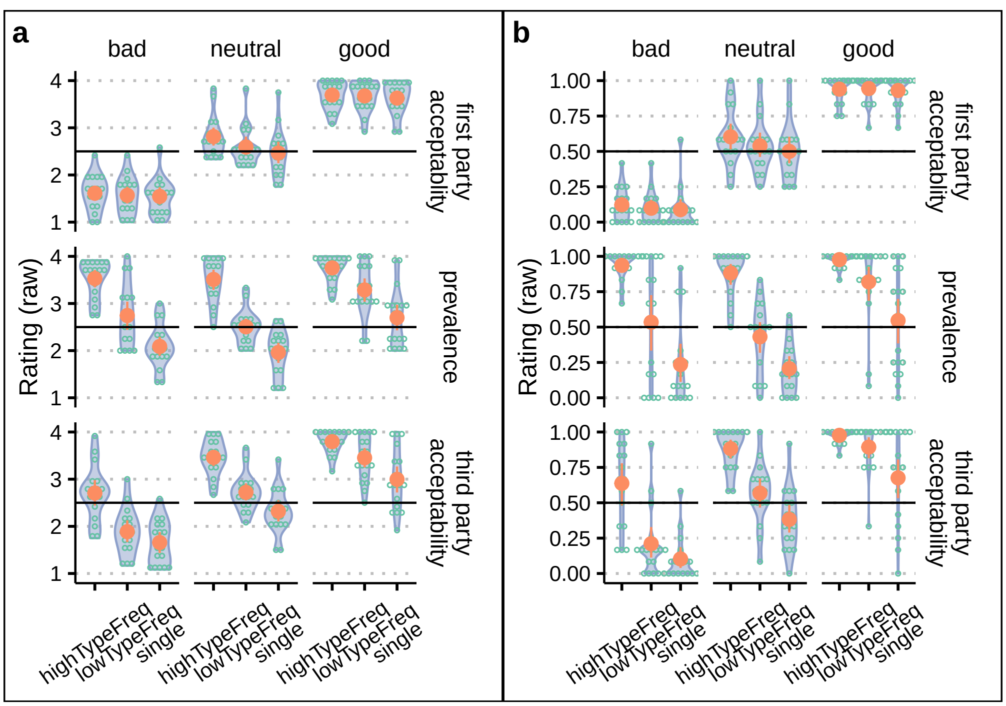

Figure 2: Ratings for City students. NOTE THAT WE HAVE SPECIFIC PLOTS FOR BINARY RATINGS BELOW. (a) Raw ratings. (b) Binary ratings.

2.2 Difference scores

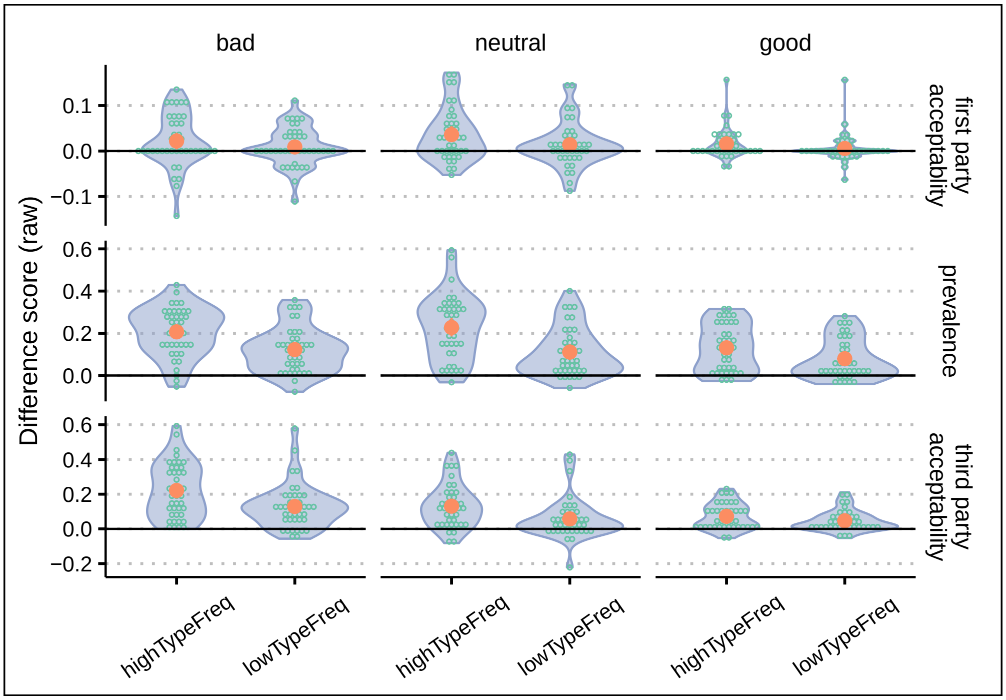

Figure 3: Raw difference scores vs. single occurrence actions for Testable participants

Table 3: Descriptive for the difference scores against the single occurrence condition using the binary as well as the untransformed data. The p value reflects a Wilcoxon test against zero.

			Binary ratings				**Raw ratings**			
Valency	Action Status	N	M	SE	Cohen's d	p	M	SE	Cohen's d	p
city - first party acceptablity										
bad	highTypeFreq	17	0.227	0.100	0.570	0.032	0.022	0.010	0.551	0.054
bad	lowTypeFreq	17	0.145	0.095	0.383	0.247	0.007	0.011	0.147	0.551
neutral	highTypeFreq	17	0.097	0.044	0.544	0.032	0.069	0.022	0.776	0.003
neutral	lowTypeFreq	17	0.045	0.036	0.314	0.299	0.029	0.014	0.530	0.045
good	highTypeFreq	18	0.006	0.007	0.205	0.349	0.010	0.004	0.659	0.014
good	lowTypeFreq	18	0.008	0.006	0.320	0.207	0.007	0.005	0.330	0.213
city - prevalence										
bad	highTypeFreq	17	0.665	0.087	1.905	0.000	0.259	0.035	1.856	0.000
bad	lowTypeFreq	17	0.312	0.096	0.809	0.010	0.133	0.030	1.120	0.001
neutral	highTypeFreq	17	0.636	0.080	1.995	0.000	0.284	0.040	1.773	0.000
neutral	lowTypeFreq	17	0.383	0.140	0.686	0.016	0.130	0.025	1.299	0.001
good	highTypeFreq	18	0.353	0.080	1.065	0.001	0.172	0.026	1.580	0.000
good	lowTypeFreq	18	0.221	0.094	0.572	0.029	0.102	0.024	1.031	0.001
city - third party acceptability										
bad	highTypeFreq	17	0.708	0.088	2.016	0.000	0.238	0.046	1.286	0.001
bad	lowTypeFreq	17	0.284	0.142	0.499	0.056	0.067	0.021	0.802	0.002
neutral	highTypeFreq	17	0.420	0.069	1.520	0.000	0.201	0.029	1.713	0.000
neutral	lowTypeFreq	17	0.216	0.081	0.666	0.010	0.084	0.021	0.997	0.002

Table 3: Descriptive for the difference scores against the single occurrence condition using the binary as well as the untransformed data. The p value reflects a Wilcoxon test against zero. *(continued)*

			Binary ratings				**Raw ratings**			
Valency	Action Status	N	M	SE	Cohen's d	p	M	SE	Cohen's d	p
good	highTypeFreq	18	0.233	0.072	0.786	0.002	0.126	0.026	1.171	0.000
good	lowTypeFreq	18	0.184	0.069	0.643	0.009	0.076	0.021	0.889	0.000
testable - first party acceptablity										
bad	highTypeFreq	41	0.050	0.087	0.090	0.587	0.022	0.009	0.382	0.024
bad	lowTypeFreq	41	-0.017	0.056	-0.049	0.968	0.009	0.007	0.211	0.165
neutral	highTypeFreq	39	0.032	0.022	0.233	0.226	0.036	0.009	0.622	0.001
neutral	lowTypeFreq	39	0.005	0.018	0.043	0.820	0.014	0.008	0.281	0.173
good	highTypeFreq	39	0.008	0.005	0.237	0.204	0.016	0.005	0.475	0.004
good	lowTypeFreq	39	0.000	0.004	-0.004	0.957	0.006	0.005	0.176	0.313
testable - prevalence										
bad	highTypeFreq	41	0.538	0.060	1.418	0.000	0.207	0.018	1.803	0.000
bad	lowTypeFreq	41	0.428	0.060	1.125	0.000	0.123	0.017	1.151	0.000
neutral	highTypeFreq	39	0.516	0.067	1.256	0.000	0.227	0.024	1.521	0.000
neutral	lowTypeFreq	39	0.379	0.066	0.932	0.000	0.111	0.018	0.977	0.000
good	highTypeFreq	39	0.224	0.047	0.771	0.000	0.131	0.018	1.176	0.000
good	lowTypeFreq	39	0.198	0.046	0.703	0.000	0.079	0.015	0.851	0.000
testable - third party acceptability										
bad	highTypeFreq	41	0.565	0.063	1.407	0.000	0.221	0.025	1.418	0.000
bad	lowTypeFreq	41	0.393	0.070	0.887	0.000	0.129	0.020	1.014	0.000
neutral	highTypeFreq	39	0.206	0.050	0.663	0.000	0.131	0.020	1.066	0.000
neutral	lowTypeFreq	39	0.117	0.050	0.381	0.023	0.058	0.019	0.485	0.002
good	highTypeFreq	39	0.074	0.021	0.561	0.001	0.072	0.012	0.970	0.000
good	lowTypeFreq	39	0.063	0.022	0.469	0.004	0.047	0.010	0.794	0.000

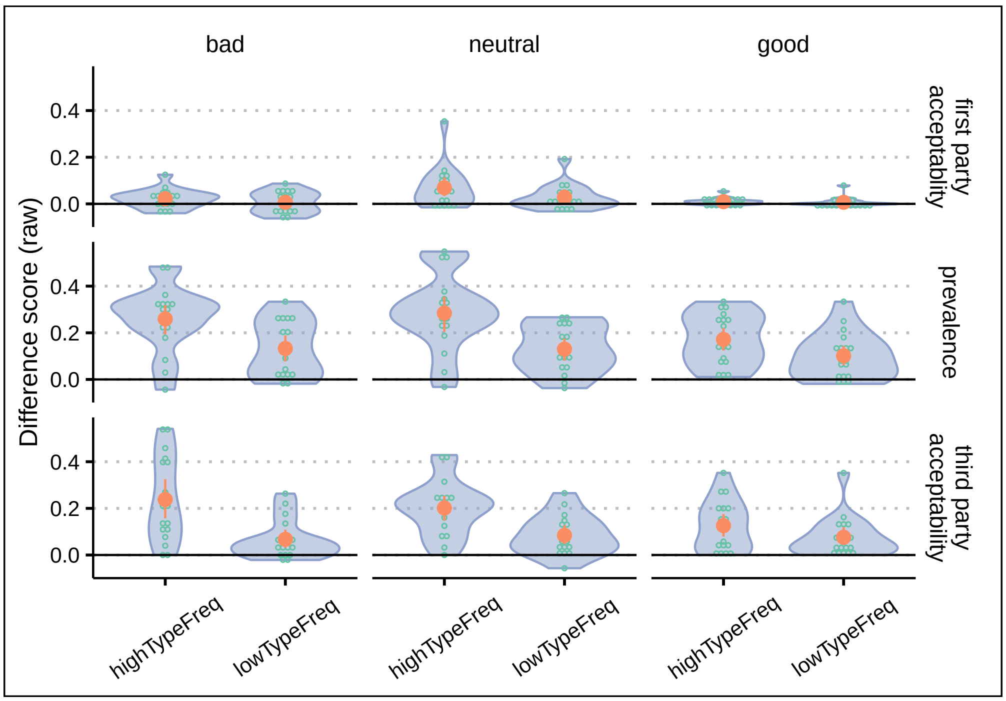

Figure 4: Raw difference scores vs. single occurrence actions for City Students

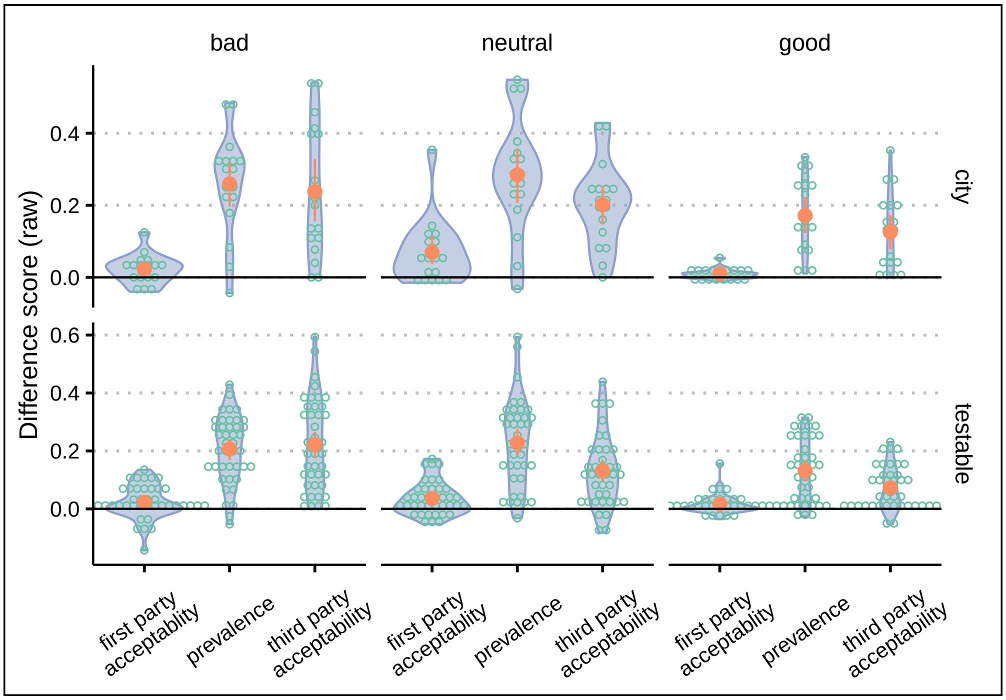

Figure 5: Raw difference scores high vs. low type frequency actions

Table 4: Descriptive for the difference scores comparing high and low type frequency conditions using the binary as well as untransformed data. The p value reflects a Wilcoxon test against zero.

		Binary ratings				Raw ratings			
Valency	N	M	SE	Cohen's d	p	M	SE	Cohen's d	p
city - first party acceptablity									
bad	17	0.100	0.072	0.345	0.092	0.022	0.010	0.551	0.054
neutral	17	0.055	0.032	0.431	0.091	0.069	0.022	0.776	0.003
good	18	-0.002	0.005	-0.095	1.000	0.010	0.004	0.659	0.014
city - prevalence									
bad	17	0.377	0.109	0.868	0.004	0.259	0.035	1.856	0.000
neutral	17	0.376	0.082	1.151	0.001	0.284	0.040	1.773	0.000
good	18	0.123	0.059	0.504	0.009	0.172	0.026	1.580	0.000
city - third party acceptability									
bad	17	0.527	0.092	1.436	0.001	0.238	0.046	1.286	0.001
neutral	17	0.234	0.058	1.007	0.001	0.201	0.029	1.713	0.000
good	18	0.054	0.031	0.429	0.052	0.126	0.026	1.171	0.000
testable - first party acceptablity									
bad	41	0.067	0.074	0.142	0.615	0.022	0.009	0.382	0.024
neutral	39	0.027	0.018	0.245	0.085	0.036	0.009	0.622	0.001
good	39	0.008	0.005	0.264	0.140	0.016	0.005	0.475	0.004
testable - prevalence									
bad	41	0.204	0.048	0.677	0.000	0.207	0.018	1.803	0.000
neutral	39	0.253	0.055	0.742	0.000	0.227	0.024	1.521	0.000
good	39	0.030	0.016	0.295	0.063	0.131	0.018	1.176	0.000

Table 4: Descriptive for the difference scores comparing high and low type frequency conditions using the binary as well as untransformed data. The p value reflects a Wilcoxon test against zero. *(continued)*

		Binary ratings				Raw ratings			
Valency	N	M	SE	Cohen's d	p	M	SE	Cohen's d	p
testable - third party acceptability									
bad	41	0.274	0.060	0.720	0.000	0.221	0.025	1.418	0.000
neutral	39	0.104	0.033	0.513	0.000	0.131	0.020	1.066	0.000
good	39	0.011	0.005	0.401	0.024	0.072	0.012	0.970	0.000

2.2.1 Prevalence (difference scores; everything is printed in Appendix)

As shown in Table 8, a number of cells deviated from normality

As shown in Table 9, we observed no main effect of Valency on the difference score against single occurrence actions for prevalence. We observed a main effect of Action Status as well as a marginal interaction between Action Status and Valency.

As shown in Table 9, separate ANOVAs for each valency revealed a strong main effect of Action Status only for all three valencies.

Finally, Table 9 shows separate ANOVAs for each Action Status. The main effect of valency was significant only for high type frequency scenarios.

2.2.1.1 Difference scores comparing high type frequency against low type frequency As shown in Table 10, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 11, we observed a marginal main effect of Valency on difference scores for first-party acceptability.

As shown in Table 11, separate ANOVAs for each valency pair revealed a strong main effect of Action Status only for the bad vs. good comparison the the neutral vs. good comparison.

2.2.2 Third-party acceptability (difference scores)

2.2.2.1 Difference scores against single actions As shown in Table 12, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 13, we observed no main effect of Valency on the difference score against single occurrence actions for first-party acceptability. , we observed a main effect of Action Status as well as an interaction between Action Status and Valency.

As shown in Table 13, separate ANOVAs for each valency revealed a strong main effect of Action Status only for neutral scenarios. The main effect was not significant for good or bad scenarios. Again, this is redundant with the direct difference scores.

Finally, Table **??** shows separate ANOVAs for each Action Status. The main effect of valency was marginal;ly significant only for high type frequency scenarios.

2.2.2.2 Difference scores comparing high type frequency against low type frequency As shown in Table 14, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 15, we observed a marginal main effect of Valency on difference scores for first-party acceptability.

As shown in Table 15, separate ANOVAs for each valency pair revealed a strong main effect of Action Status only for the bad vs. good comparison, and a marginal one for the neutral vs. good comparison.

Table 5: ANOVA for difference scores comparing the high vs low type frequency conditions for first-person acceptability. (Top) Overall ANOVA. (Bottom) Differences between frequency conditions separately for each valency.

myValence	Effect	df1	df2	F	p	η_p^2
city - Overall						
	myValence	2	49	5.310	0.008	0.178
testable - Overall						
	myValence	2	116	1.590	0.209	0.027
testable - By valency						
neutral vs. good	myValence	1	76	3.520	0.064	0.044
neutral vs. good	myValence	1	33	7.810	0.009	0.191
bad vs. good	myValence	1	78	0.332	0.566	0.004
city - By valency						
bad vs. good	myValence	1	33	1.540	0.223	0.045
bad vs. neutral	myValence	1	78	1.170	0.282	0.015
bad vs. neutral	myValence	1	32	3.900	0.057	0.109

2.2.3 First-party acceptability (difference scores)

2.2.3.1 Difference scores against single actions As shown in Table 16, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 17, we observed the expected main effect of Valency on first-party acceptability. Further, we observed a main effect of Action Status as well as an interaction between Action Status and Valency. (Not that including the factor Action Status is redundant with the direct difference score below.)

As shown in Table 17, separate ANOVAs for each valency revealed a strong main effect of Action Status only for neutral scenarios. The main effect was not significant for good or bad scenarios. Again, this is redundant with the direct difference scores.

Finally, Table 17 shows separate ANOVAs for each Action Status. The main effect of valency was significant only for high type frequency scenarios.

2.2.3.2 Difference scores comparing high type frequency against low type frequency As shown in Table 18, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table **??**, we observed the expected main effect of Valency on difference scores for first-party acceptability.

As shown in Table 19, separate ANOVAs for each valency revealed a strong main effect of Action Status only for neutral scenarios. The main effect was not significant for good or bad scenarios. Again, this is redundant with the direct difference scores.

3 Binary ratings (analyzed using GLMER)

We analyze the binary ratings using two types of GLMMs. For valency, we set the reference level to neutral. For the frequency condition, we fit a GLMMM including all frequency conditions and setting the reference level to single occurrence events (corresponding to the first type of difference scores); we also fit a GLMM excluding the single ocurrence events (corresponding to the second type of differences scores).

3.1 Ratings

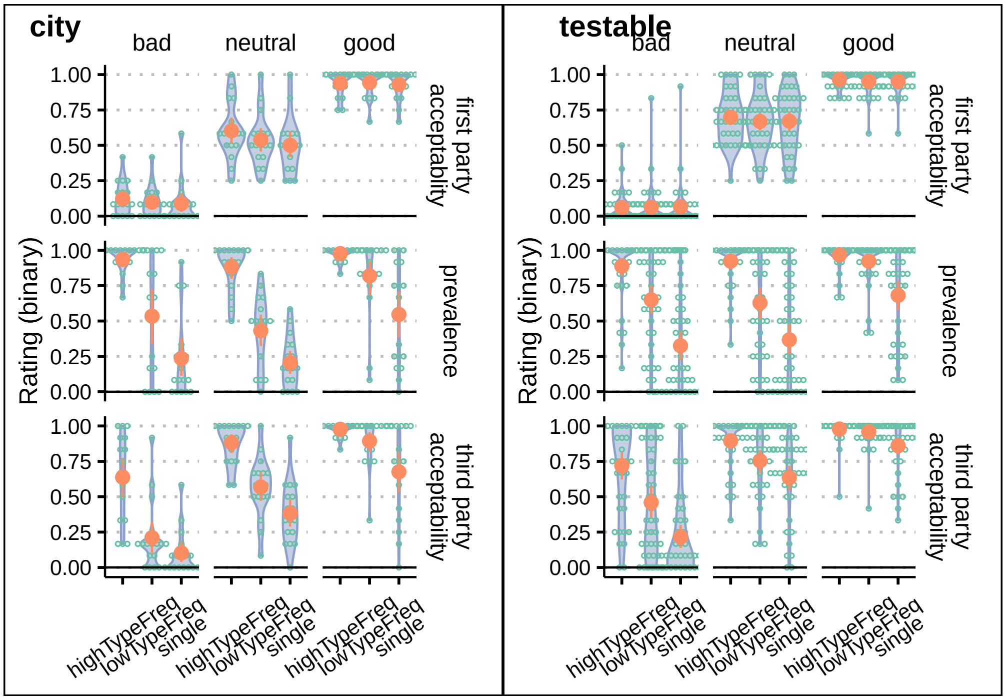

Figure 6: Bin ratings

3.2 Overall (GLM)

3.2.1 Prevalence (GLMM)

```
## $city
## Data: dat.social.type.token.sequential.for.glmm.3freq.cond %>% dplyr::filter(population ==  ...
## Models:
## .y: rating ~ myQueriedActionStatus + myValence + (1 | filename) + (1 | myQueriedAction)
## .x: rating ~ myQueriedActionStatus * myValence + (1 | filename) + (1 | myQueriedAction)
##    npar  AIC  BIC logLik deviance Chisq Df Pr(>Chisq)
## .y    7 1606 1644   -796     1592
## .x   11 1605 1666   -791     1583  8.56  4      0.073 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## $testable
## Data: dat.social.type.token.sequential.for.glmm.3freq.cond %>% dplyr::filter(population ==  ...
## Models:
## .y: rating ~ myQueriedActionStatus + myValence + (1 | filename) + (1 | myQueriedAction)
## .x: rating ~ myQueriedActionStatus * myValence + (1 | filename) + (1 | myQueriedAction)
```

```
##    npar  AIC  BIC logLik deviance Chisq Df Pr(>Chisq)
## .y    7 3244 3288  -1615     3230
## .x   11 3236 3306  -1607     3214  15.4  4     0.0039 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3.2.2 Third-party acceptability

3.2.3 First-party acceptability

3.3 Difference scores (GLMER or ANOVA?)

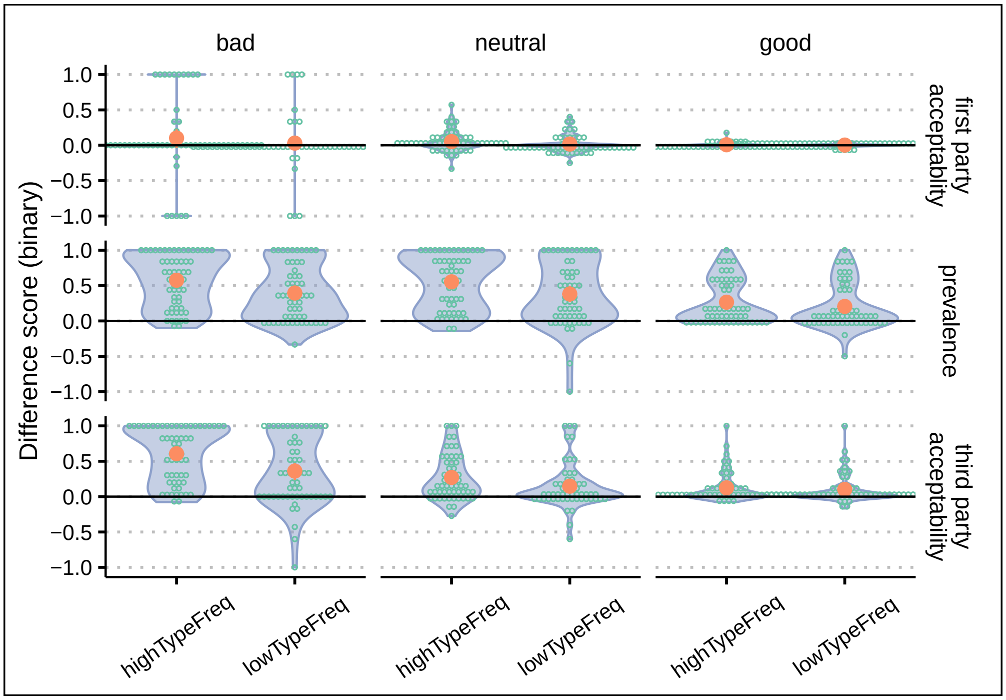

Figure 7: Binary difference scores vs. single occurrence actions

3.3.1 Prevalence

3.3.2 Third-party acceptability

3.3.3 First-party acceptability

Table 6: GLMER results, using the full models comparing the three frequency conditions. ONLY EFFECTS < .1 ARE SHOWN; CHANGE THIS

	Log-odds			Odd ratios				
term	Estimate	SE	CI	Estimate	SE	CI	t	p
city - First party acceptability								
myQueriedActionStatushighTypeFreq	0.793	0.279	[0.246, 1.34]	2.211	0.617	[1.28, 3.82]	2.84	0.004
myValencebad	-4.087	1.082	[-6.21, -1.97]	0.017	0.018	[0.00201, 0.14]	-3.78	0.000
myValencegood	4.853	1.141	[2.62, 7.09]	128.135	146.240	[13.7, 1200]	4.25	0.000
city - Overall								
questionthird.party.acceptability	-0.583	0.221	[-1.02, -0.149]	0.558	0.124	[0.362, 0.861]	-2.63	0.008
questionprevalence	-1.620	0.245	[-2.1, -1.14]	0.198	0.048	[0.123, 0.32]	-6.63	0.000
myQueriedActionStatushighTypeFreq	0.507	0.220	[0.0751, 0.939]	1.660	0.366	[1.08, 2.56]	2.30	0.021
myValencebad	-2.698	0.523	[-3.72, -1.67]	0.067	0.035	[0.0242, 0.188]	-5.16	0.000
myValencegood	3.311	0.538	[2.26, 4.37]	27.425	14.744	[9.56, 78.7]	6.16	0.000
questionthird.party.acceptability:myQueriedActionStatuslowTypeFreq	0.728	0.312	[0.117, 1.34]	2.071	0.645	[1.12, 3.81]	2.33	0.020
questionprevalence:myQueriedActionStatuslowTypeFreq	1.093	0.328	[0.45, 1.73]	2.982	0.977	[1.57, 5.67]	3.33	0.001
questionthird.party.acceptability:myQueriedActionStatushighTypeFreq	2.426	0.356	[1.73, 3.13]	11.318	4.034	[5.63, 22.8]	6.81	0.000
questionprevalence:myQueriedActionStatushighTypeFreq	3.463	0.372	[2.73, 4.19]	31.916	11.888	[15.4, 66.2]	9.30	0.000
questionthird.party.acceptability:myValencebad	0.771	0.417	[-0.0453, 1.59]	2.162	0.901	[0.956, 4.89]	1.85	0.064
questionprevalence:myValencebad	2.933	0.398	[2.15, 3.71]	18.777	7.474	[8.61, 41]	7.37	0.000
questionthird.party.acceptability:myValencegood	-1.676	0.397	[-2.45, -0.897]	0.187	0.074	[0.0859, 0.408]	-4.22	0.000
questionprevalence:myValencegood	-1.388	0.411	[-2.19, -0.582]	0.250	0.103	[0.111, 0.559]	-3.37	0.001
city - Prevalence								
(Intercept)	-1.522	0.376	[-2.26, -0.785]	0.218	0.082	[0.104, 0.456]	-4.05	0.000
myQueriedActionStatuslowTypeFreq	1.203	0.238	[0.737, 1.67]	3.329	0.791	[2.09, 5.3]	5.06	0.000
myQueriedActionStatushighTypeFreq	3.766	0.306	[3.17, 4.36]	43.212	13.203	[23.7, 78.6]	12.33	0.000
myValencegood	1.864	0.524	[0.838, 2.89]	6.450	3.378	[2.31, 18]	3.56	0.000
myQueriedActionStatuslowTypeFreq:myValencebad	0.643	0.359	[-0.0602, 1.35]	1.902	0.682	[0.942, 3.84]	1.79	0.073
myQueriedActionStatushighTypeFreq:myValencebad	1.335	0.513	[0.33, 2.34]	3.800	1.949	[1.39, 10.4]	2.60	0.009
myQueriedActionStatuslowTypeFreq:myValencegood	0.632	0.360	[-0.0732, 1.34]	1.882	0.677	[0.929, 3.81]	1.76	0.079
city - Third party acceptability								
myQueriedActionStatuslowTypeFreq	0.906	0.222	[0.47, 1.34]	2.473	0.549	[1.6, 3.82]	4.08	0.000
myQueriedActionStatushighTypeFreq	2.876	0.286	[2.32, 3.44]	17.743	5.076	[10.1, 31.1]	10.05	0.000
myValencebad	-2.046	0.517	[-3.06, -1.03]	0.129	0.067	[0.047, 0.356]	-3.96	0.000
myValencegood	1.583	0.481	[0.64, 2.53]	4.872	2.344	[1.9, 12.5]	3.29	0.001
myQueriedActionStatuslowTypeFreq:myValencegood	0.831	0.368	[0.11, 1.55]	2.297	0.846	[1.12, 4.73]	2.26	0.024
testable - First party acceptability								
(Intercept)	1.468	0.573	[0.344, 2.59]	4.342	2.490	[1.41, 13.4]	2.56	0.010
myValencebad	-6.070	0.868	[-7.77, -4.37]	0.002	0.002	[0.000422, 0.0127]	-7.00	0.000
myValencegood	3.622	0.886	[1.89, 5.36]	37.427	33.149	[6.6, 212]	4.09	0.000
testable - Overall								
(Intercept)	1.003	0.308	[0.399, 1.61]	2.726	0.840	[1.49, 4.99]	3.25	0.001
questionprevalence	-1.770	0.165	[-2.09, -1.45]	0.170	0.028	[0.123, 0.235]	-10.74	0.000
myValencebad	-4.251	0.462	[-5.16, -3.35]	0.014	0.007	[0.00576, 0.0352]	-9.20	0.000
myValencegood	2.852	0.484	[1.9, 3.8]	17.325	8.378	[6.72, 44.7]	5.90	0.000
questionthird.party.acceptability:myQueriedActionStatuslowTypeFreq	0.797	0.235	[0.337, 1.26]	2.219	0.521	[1.4, 3.52]	3.40	0.001
questionprevalence:myQueriedActionStatuslowTypeFreq	1.521	0.230	[1.07, 1.97]	4.576	1.053	[2.92, 7.18]	6.61	0.000
questionthird.party.acceptability:myQueriedActionStatushighTypeFreq	1.921	0.262	[1.41, 2.43]	6.827	1.788	[4.09, 11.4]	7.33	0.000
questionprevalence:myQueriedActionStatushighTypeFreq	3.872	0.281	[3.32, 4.42]	48.053	13.515	[27.7, 83.4]	13.77	0.000
questionthird.party.acceptability:myValencebad	1.825	0.280	[1.28, 2.37]	6.202	1.738	[3.58, 10.7]	6.51	0.000
questionprevalence:myValencebad	4.075	0.278	[3.53, 4.62]	58.847	16.354	[34.1, 101]	14.66	0.000
questionthird.party.acceptability:myValencegood	-1.154	0.315	[-1.77, -0.537]	0.315	0.099	[0.17, 0.584]	-3.67	0.000
questionprevalence:myValencegood	-0.973	0.306	[-1.57, -0.374]	0.378	0.116	[0.207, 0.688]	-3.18	0.001
questionthird.party.acceptability:myQueriedActionStatuslowTypeFreq:myValencebad	0.674	0.395	[-0.0996, 1.45]	1.963	0.775	[0.905, 4.26]	1.71	0.088
questionthird.party.acceptability:myQueriedActionStatushighTypeFreq:myValencebad	1.013	0.417	[0.195, 1.83]	2.755	1.150	[1.22, 6.24]	2.43	0.015
questionprevalence:myQueriedActionStatushighTypeFreq:myValencegood	-1.057	0.535	[-2.1, -0.00858]	0.348	0.186	[0.122, 0.991]	-1.98	0.048
testable - Prevalence								
(Intercept)	-0.781	0.347	[-1.46, -0.101]	0.458	0.159	[0.232, 0.904]	-2.25	0.024
myQueriedActionStatuslowTypeFreq	1.723	0.182	[1.37, 2.08]	5.599	1.016	[3.92, 7.99]	9.49	0.000
myQueriedActionStatushighTypeFreq	4.340	0.259	[3.83, 4.85]	76.680	19.874	[46.1, 127]	16.74	0.000
myValencegood	2.137	0.496	[1.16, 3.11]	8.473	4.202	[3.21, 22.4]	4.31	0.000
myQueriedActionStatuslowTypeFreq:myValencegood	0.569	0.292	[-0.00335, 1.14]	1.767	0.516	[0.997, 3.13]	1.95	0.051
myQueriedActionStatushighTypeFreq:myValencegood	-1.015	0.399	[-1.8, -0.234]	0.362	0.144	[0.166, 0.791]	-2.55	0.011
testable - Third party acceptability								
(Intercept)	0.872	0.359	[0.168, 1.58]	2.391	0.858	[1.18, 4.83]	2.43	0.015
myQueriedActionStatuslowTypeFreq	0.844	0.174	[0.502, 1.19]	2.325	0.405	[1.65, 3.27]	4.84	0.000
myQueriedActionStatushighTypeFreq	2.244	0.216	[1.82, 2.67]	9.427	2.040	[6.17, 14.4]	10.37	0.000
myValencebad	-2.827	0.511	[-3.83, -1.83]	0.059	0.030	[0.0217, 0.161]	-5.53	0.000
myValencegood	1.819	0.524	[0.792, 2.85]	6.168	3.234	[2.21, 17.2]	3.47	0.001
myQueriedActionStatuslowTypeFreq:myValencebad	0.887	0.253	[0.392, 1.38]	2.429	0.613	[1.48, 3.98]	3.51	0.000
myQueriedActionStatushighTypeFreq:myValencebad	1.173	0.298	[0.589, 1.76]	3.231	0.962	[1.8, 5.79]	3.94	0.000
myQueriedActionStatuslowTypeFreq:myValencegood	0.829	0.336	[0.171, 1.49]	2.291	0.769	[1.19, 4.42]	2.47	0.014

Table 7: GLMER results, using the full models comparing the two frequency conditions. ONLY EFFECTS < .1 ARE SHOWN; CHANGE THIS

	Log-odds			Odd ratios				
term	Estimate	SE	CI	Estimate	SE	CI	t	p
city - First party acceptability								
myQueriedActionStatushighTypeFreq	0.490	0.277	[-0.0526, 1.03]	1.632	0.451	[0.949, 2.81]	1.77	0.077
myValencebad	-4.110	1.044	[-6.16, -2.06]	0.016	0.017	[0.00212, 0.127]	-3.94	0.000
myValencegood	4.580	1.123	[2.38, 6.78]	97.517	109.511	[10.8, 881]	4.08	0.000
city - Overall								
questionprevalence	-0.522	0.219	[-0.951, -0.094]	0.593	0.130	[0.386, 0.91]	-2.39	0.017
myValencebad	-2.734	0.477	[-3.67, -1.8]	0.065	0.031	[0.0255, 0.165]	-5.73	0.000
myValencegood	3.162	0.513	[2.16, 4.17]	23.614	12.109	[8.64, 64.5]	6.17	0.000
questionthird.party.acceptability:myQueriedActionStatushighTypeFreq	1.684	0.352	[0.994, 2.37]	5.386	1.896	[2.7, 10.7]	4.78	0.000
questionprevalence:myQueriedActionStatushighTypeFreq	2.349	0.354	[1.66, 3.04]	10.475	3.709	[5.23, 21]	6.63	0.000
questionthird.party.acceptability:myValencebad	0.867	0.377	[0.128, 1.61]	2.379	0.897	[1.14, 4.98]	2.30	0.022
questionprevalence:myValencebad	3.237	0.368	[2.52, 3.96]	25.457	9.374	[12.4, 52.4]	8.79	0.000
questionthird.party.acceptability:myValencegood	-0.937	0.439	[-1.8, -0.0777]	0.392	0.172	[0.166, 0.925]	-2.14	0.033
questionprevalence:myValencegood	-0.994	0.420	[-1.82, -0.17]	0.370	0.156	[0.162, 0.844]	-2.36	0.018
city - Prevalence								
myQueriedActionStatushighTypeFreq	2.631	0.288	[2.07, 3.19]	13.886	3.993	[7.9, 24.4]	9.15	0.000
myValencegood	2.512	0.596	[1.34, 3.68]	12.327	7.344	[3.84, 39.6]	4.22	0.000
city - Third party acceptability								
myQueriedActionStatushighTypeFreq	1.976	0.281	[1.42, 2.53]	7.213	2.028	[4.16, 12.5]	7.03	0.000
myValencebad	-1.983	0.468	[-2.9, -1.07]	0.138	0.064	[0.055, 0.345]	-4.24	0.000
myValencegood	2.262	0.499	[1.28, 3.24]	9.598	4.787	[3.61, 25.5]	4.54	0.000
testable - First party acceptability								
(Intercept)	1.483	0.609	[0.289, 2.68]	4.405	2.683	[1.33, 14.5]	2.43	0.015
myValencebad	-6.194	0.934	[-8.02, -4.36]	0.002	0.002	[0.000328, 0.0127]	-6.63	0.000
myValencegood	3.908	0.963	[2.02, 5.8]	49.816	47.979	[7.54, 329]	4.06	0.000
testable - Overall								
(Intercept)	1.037	0.344	[0.363, 1.71]	2.821	0.970	[1.44, 5.54]	3.02	0.003
questionthird.party.acceptability	0.598	0.173	[0.259, 0.936]	1.818	0.314	[1.3, 2.55]	3.46	0.001
myValencebad	-4.443	0.512	[-5.45, -3.44]	0.012	0.006	[0.00431, 0.0321]	-8.67	0.000
myValencegood	3.170	0.540	[2.11, 4.23]	23.809	12.859	[8.26, 68.6]	5.87	0.000
questionthird.party.acceptability:myQueriedActionStatushighTypeFreq	1.168	0.270	[0.638, 1.7]	3.214	0.869	[1.89, 5.46]	4.32	0.000
questionprevalence:myQueriedActionStatushighTypeFreq	2.444	0.283	[1.89, 3]	11.524	3.260	[6.62, 20.1]	8.64	0.000
questionthird.party.acceptability:myValencebad	2.573	0.289	[2.01, 3.14]	13.110	3.794	[7.43, 23.1]	8.89	0.000
questionprevalence:myValencebad	4.511	0.292	[3.94, 5.08]	91.043	26.576	[51.4, 161]	15.46	0.000
questionprevalence:myQueriedActionStatushighTypeFreq:myValencegood	-1.709	0.558	[-2.8, -0.616]	0.181	0.101	[0.0607, 0.54]	-3.06	0.002
testable - Prevalence								
(Intercept)	1.200	0.418	[0.38, 2.02]	3.320	1.389	[1.46, 7.54]	2.87	0.004
myQueriedActionStatushighTypeFreq	2.768	0.248	[2.28, 3.25]	15.921	3.943	[9.8, 25.9]	11.18	0.000
myValencegood	2.965	0.642	[1.71, 4.22]	19.402	12.461	[5.51, 68.3]	4.62	0.000
myQueriedActionStatushighTypeFreq:myValencegood	-1.630	0.426	[-2.47, -0.794]	0.196	0.084	[0.0849, 0.452]	-3.82	0.000
testable - Third party acceptability								
(Intercept)	2.004	0.457	[1.11, 2.9]	7.420	3.393	[3.03, 18.2]	4.38	0.000
myQueriedActionStatushighTypeFreq	1.546	0.235	[1.08, 2.01]	4.693	1.104	[2.96, 7.44]	6.57	0.000
myValencebad	-2.214	0.627	[-3.44, -0.985]	0.109	0.069	[0.032, 0.373]	-3.53	0.000
myValencegood	3.213	0.734	[1.77, 4.65]	24.857	18.244	[5.9, 105]	4.38	0.000

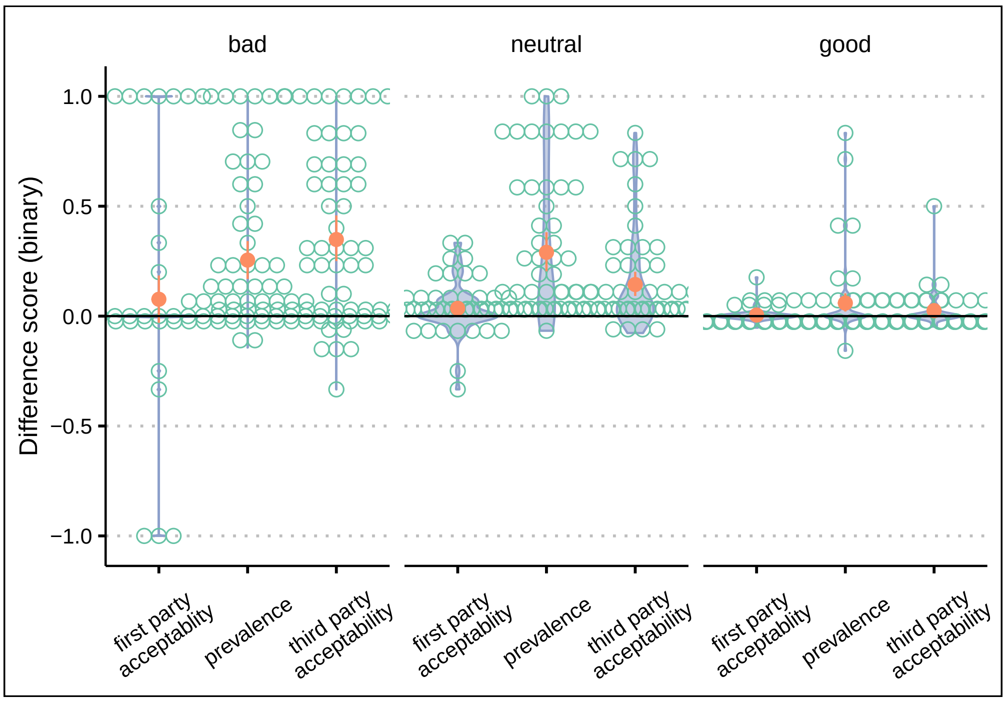

Figure 8: Binary difference scores high vs. low type frequency actions

Table 8: SW for difference scores vs. single occurrence actions for untranformed (raw) prevalence ratings.

myValence	myQueriedActionStatus	population	W	p.value	p<=.05
bad	highTypeFreq	city	0.939	0.307	
neutral	highTypeFreq	city	0.955	0.538	
good	highTypeFreq	city	0.929	0.190	
bad	lowTypeFreq	city	0.900	0.067	.
neutral	lowTypeFreq	city	0.936	0.269	
good	lowTypeFreq	city	0.929	0.187	
bad	highTypeFreq	testable	0.973	0.432	
neutral	highTypeFreq	testable	0.957	0.139	
good	highTypeFreq	testable	0.920	0.008	**
bad	lowTypeFreq	testable	0.953	0.089	.
neutral	lowTypeFreq	testable	0.910	0.004	**
good	lowTypeFreq	testable	0.895	0.002	**

4 Appendix

4.1 ANOVAS for difference scores for raw data

4.1.1 Prevalence (difference scores; everything is printed in Appendix)

As shown in Table 8, a number of cells deviated from normality

As shown in Table 9, we observed no main effect of Valency on the difference score against single occurrence actions for prevalence. We observed a main effect of Action Status as well as a marginal interaction between Action Status and Valency.

As shown in Table 9, separate ANOVAs for each valency revealed a strong main effect of Action Status only for all three valencies.

Finally, Table 9 shows separate ANOVAs for each Action Status. The main effect of valency was significant only for high type frequency scenarios.

4.1.1.1 Difference scores comparing high type frequency against low type frequency As shown in Table 10, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 11, we observed a marginal main effect of Valency on difference scores for first-party acceptability.

As shown in Table 11, separate ANOVAs for each valency pair revealed a strong main effect of Action Status only for the bad vs. good comparison the the neutral vs. good comparison.

Table 9: ANOVA for difference scores comparing each frequency condition to the single occurrence condition for untransformed prevalence ratings. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency . (Bottom). Differences for each frequency conditions separately.

Effect	df1	df2	F	p	η_p^2
city - Overall					
myValence	2	49	2.090	0.134	0.079
myQueriedActionStatus	1	49	60.800	0.000	0.527
myQueriedActionStatus:myValence	2	49	2.810	0.070	0.049
testable - Overall					
myValence	2	116	4.560	0.012	0.073
myQueriedActionStatus	1	116	84.700	0.000	0.406
myQueriedActionStatus:myValence	2	116	4.110	0.019	0.039
city - Follow-up of valency - bad					
myQueriedActionStatus	1	16	15.100	0.001	0.486
city - Follow-up of valency - neutral					
myQueriedActionStatus	1	16	40.000	0.000	0.714
city - Follow-up of valency - good					
myQueriedActionStatus	1	17	13.000	0.002	0.434
testable - Follow-up of valency - bad					
myQueriedActionStatus	1	40	25.500	0.000	0.389
testable - Follow-up of valency - neutral					
myQueriedActionStatus	1	38	39.700	0.000	0.511
testable - Follow-up of valency - good					
myQueriedActionStatus	1	38	20.800	0.000	0.354
city - Follow-up of Action Status - highTypeFreq					
myValence	2	49	3.280	0.046	0.118
city - Follow-up of Action Status - lowTypeFreq					
myValence	2	49	0.469	0.628	0.019
testable - Follow-up of Action Status - highTypeFreq					
myValence	2	116	6.330	0.002	0.098
testable - Follow-up of Action Status - lowTypeFreq					
myValence	2	116	1.800	0.170	0.030

Table 10: SW acceptability for raw prevalence ratings for difference scores comparing high vs. low type frequency conditions

myValence	population	W	p.value	p<=.05
bad	city	0.910	0.001	***
neutral	city	0.944	0.018	*
good	city	0.842	0.000	***
bad	testable	0.951	0.000	***
neutral	testable	0.920	0.000	***
good	testable	0.861	0.000	***

Table 11: ANOVA for difference scores comparing the high vs low type frequency conditions for raw prevalence ratings. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency. (Bottom). Differences for each frequency condition separately.

myValence	Effect	df1	df2	F	p	η_p^2
city - Overall						
	myValence	2	49	3.280	0.046	0.118
testable - Overall						
	myValence	2	116	6.330	0.002	0.098
city - By valency						
bad vs. good	myValence	1	33	4.340	0.045	0.116
bad vs. neutral	myValence	1	32	0.232	0.633	0.007
neutral vs. good	myValence	1	33	5.990	0.020	0.154
testable - By valency						
bad vs. good	myValence	1	78	8.950	0.004	0.103
bad vs. neutral	myValence	1	78	0.489	0.486	0.006
neutral vs. good	myValence	1	76	10.400	0.002	0.121

Table 12: SW acceptability for difference scores against single occurrence conditions for third party acceptability.

myValence	myQueriedActionStatus	population	W	p.value	p<=.05
bad	highTypeFreq	city	0.914	0.116	
neutral	highTypeFreq	city	0.961	0.659	
good	highTypeFreq	city	0.921	0.136	
bad	lowTypeFreq	city	0.858	0.014	*
neutral	lowTypeFreq	city	0.969	0.801	
good	lowTypeFreq	city	0.780	0.001	***
bad	highTypeFreq	testable	0.950	0.071	.
neutral	highTypeFreq	testable	0.959	0.165	
good	highTypeFreq	testable	0.935	0.025	*
bad	lowTypeFreq	testable	0.888	0.001	***
neutral	lowTypeFreq	testable	0.833	0.000	***
good	lowTypeFreq	testable	0.911	0.005	**

4.1.2 Third-party acceptability (difference scores)

4.1.2.1 Difference scores against single actions As shown in Table 12, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 13, we observed no main effect of Valency on the difference score against single occurrence actions for first-party acceptability. , we observed a main effect of Action Status as well as an interaction between Action Status and Valency.

As shown in Table 13, separate ANOVAs for each valency revealed a strong main effect of Action Status only for neutral scenarios. The main effect was not significant for good or bad scenarios. Again, this is redundant with the direct difference scores.

Finally, Table **??** shows separate ANOVAs for each Action Status. The main effect of valency was marginal;ly significant only for high type frequency scenarios.

4.1.2.2 Difference scores comparing high type frequency against low type frequency As shown in Table 14, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 15, we observed a marginal main effect of Valency on difference scores for first-party acceptability.

As shown in Table 15, separate ANOVAs for each valency pair revealed a strong main effect of Action Status only for the bad vs. good comparison, and a marginal one for the neutral vs. good comparison.

Table 13: ANOVA for difference scores comparing each frequency condition to the single occurrence condition for raw third-person acceptability ratings. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency. (Bottom). Differences for each frequency conditions separately.

Effect	df1	df2	F	p	η_p^2
city - Overall					
myValence	2	49	1.280	0.288	0.050
myQueriedActionStatus	1	49	52.100	0.000	0.467
myQueriedActionStatus:myValence	2	49	5.170	0.009	0.093
testable - Overall					
myValence	2	116	12.800	0.000	0.181
myQueriedActionStatus	1	116	54.900	0.000	0.303
myQueriedActionStatus:myValence	2	116	5.220	0.007	0.058
testable - Follow-up of valency - good					
myQueriedActionStatus	1	38	12.600	0.001	0.249
testable - Follow-up of valency - neutral					
myQueriedActionStatus	1	38	27.300	0.000	0.418
testable - Follow-up of valency - bad					
myQueriedActionStatus	1	40	20.900	0.000	0.343
city - Follow-up of valency - bad					
myQueriedActionStatus	1	16	18.500	0.001	0.537
city - Follow-up of valency - neutral					
myQueriedActionStatus	1	16	37.500	0.000	0.701
city - Follow-up of valency - good					
myQueriedActionStatus	1	17	9.520	0.007	0.359
city - Follow-up of Action Status - highTypeFreq					
myValence	2	49	2.900	0.064	0.106
city - Follow-up of Action Status - lowTypeFreq					
myValence	2	49	0.174	0.841	0.007
testable - Follow-up of Action Status - highTypeFreq					
myValence	2	116	14.900	0.000	0.204
testable - Follow-up of Action Status - lowTypeFreq					
myValence	2	116	7.100	0.001	0.109

Table 14: SW for difference scores comparing high vs. low difference type frequency conditions for third party acceptability

myValence	population	W	p.value	p<=.05
bad	city	0.910	0.001	***
neutral	city	0.944	0.018	*
good	city	0.842	0.000	***
bad	testable	0.951	0.000	***
neutral	testable	0.920	0.000	***
good	testable	0.861	0.000	***

Table 15: ANOVA for difference scores comparing the high vs low type frequency conditions for third-person acceptability. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency. (Bottom). Differences for each frequency condition.

myValence	Effect	df1	df2	F	p	η_p^2
city - Overall						
	myValence	2	49	2.900	0.064	0.106
testable - Overall						
	myValence	2	116	14.900	0.000	0.204
testable - By valency						
neutral vs. good	myValence	1	76	6.460	0.013	0.078
neutral vs. good	myValence	1	33	3.930	0.056	0.106
bad vs. good	myValence	1	78	29.100	0.000	0.272
city - By valency						
bad vs. good	myValence	1	33	4.850	0.035	0.128
bad vs. neutral	myValence	1	78	8.210	0.005	0.095
bad vs. neutral	myValence	1	32	0.465	0.500	0.014

Table 16: SW for difference scores against single occurrence actions for first party acceptability.

myValence	myQueriedActionStatus	population	W	p.value	p<=.05
bad	highTypeFreq	city	0.941	0.335	
neutral	highTypeFreq	city	0.776	0.001	***
good	highTypeFreq	city	0.821	0.003	**
bad	lowTypeFreq	city	0.946	0.391	
neutral	lowTypeFreq	city	0.831	0.006	**
good	lowTypeFreq	city	0.679	0.000	***
bad	highTypeFreq	testable	0.932	0.016	*
neutral	highTypeFreq	testable	0.923	0.011	*
good	highTypeFreq	testable	0.790	0.000	***
bad	lowTypeFreq	testable	0.943	0.040	*
neutral	lowTypeFreq	testable	0.926	0.014	*
good	lowTypeFreq	testable	0.693	0.000	***

4.1.3 First-party acceptability (difference scores)

4.1.3.1 Difference scores against single actions As shown in Table 16, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 17, we observed the expected main effect of Valency on first-party acceptability. Further, we observed a main effect of Action Status as well as an interaction between Action Status and Valency. (Not that including the factor Action Status is redundant with the direct difference score below.)

As shown in Table 17, separate ANOVAs for each valency revealed a strong main effect of Action Status only for neutral scenarios. The main effect was not significant for good or bad scenarios. Again, this is redundant with the direct difference scores.

Finally, Table 17 shows separate ANOVAs for each Action Status. The main effect of valency was significant only for high type frequency scenarios.

4.1.3.2 Difference scores comparing high type frequency against low type frequency As shown in Table 18, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table **??**, we observed the expected main effect of Valency on difference scores for first-party acceptability.

As shown in Table 19, separate ANOVAs for each valency revealed a strong main effect of Action Status only for neutral scenarios. The main effect was not significant for good or bad scenarios. Again, this is redundant with the direct difference scores.

Table 17: ANOVA for difference scores comparing each frequency condition to the single occurrence condition for first-person acceptability. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency f. (Bottom). Differences for each frequency condition.

Effect	df1	df2	F	p	η_p^2
city - Overall					
myValence	2	49	4.100	0.023	0.143
myQueriedActionStatus	1	49	10.400	0.002	0.158
myQueriedActionStatus:myValence	2	49	3.430	0.040	0.103
testable - Overall					
myValence	2	116	1.150	0.321	0.019
myQueriedActionStatus	1	116	16.600	0.000	0.123
myQueriedActionStatus:myValence	2	116	0.916	0.403	0.014
testable - Follow-up of valency - good					
myQueriedActionStatus	1	38	11.300	0.002	0.230
testable - Follow-up of valency - neutral					
myQueriedActionStatus	1	38	13.400	0.001	0.261
testable - Follow-up of valency - bad					
myQueriedActionStatus	1	40	2.280	0.139	0.054
city - Follow-up of valency - bad					
myQueriedActionStatus	1	16	1.610	0.223	0.091
city - Follow-up of valency - neutral					
myQueriedActionStatus	1	16	9.450	0.007	0.371
city - Follow-up of valency - good					
myQueriedActionStatus	1	17	0.937	0.347	0.052
city - Follow-up of Action Status - highTypeFreq					
myValence	2	49	5.310	0.008	0.178
city - Follow-up of Action Status - lowTypeFreq					
myValence	2	49	1.560	0.220	0.060
testable - Follow-up of Action Status - highTypeFreq					
myValence	2	116	1.590	0.209	0.027
testable - Follow-up of Action Status - lowTypeFreq					
myValence	2	116	0.384	0.682	0.007

Table 18: SW acceptability for participants themselves for high vs. low difference scores

myValence	population	W	p.value	p<=.05
bad	city	0.910	0.001	***
neutral	city	0.944	0.018	*
good	city	0.842	0.000	***
bad	testable	0.951	0.000	***
neutral	testable	0.920	0.000	***
good	testable	0.861	0.000	***

Table 19: ANOVA for difference scores comparing the high vs low type frequency conditions for first-person acceptability. (Top) Overall ANOVA. (Bottom) Differences between frequency conditions separately for each valency.

myValence	Effect	df1	df2	F	p	η_p^2
city - Overall						
	myValence	2	49	5.310	0.008	0.178
testable - Overall						
	myValence	2	116	1.590	0.209	0.027
testable - By valency						
neutral vs. good	myValence	1	76	3.520	0.064	0.044
neutral vs. good	myValence	1	33	7.810	0.009	0.191
bad vs. good	myValence	1	78	0.332	0.566	0.004
city - By valency						
bad vs. good	myValence	1	33	1.540	0.223	0.045
bad vs. neutral	myValence	1	78	1.170	0.282	0.015
bad vs. neutral	myValence	1	32	3.900	0.057	0.109

5 Unused

5.1 Power analysis (REMOVE IN PAPER)

We use the SIMR package for power analysis [Green and MacLeod, 2016].

See https://besjournals.onlinelibrary.wiley.com/doi/full/10.1111/2041-210X.12504 for usage of extend and powerCurve

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